Comparing Scientific ML Methods for Nonlinear ODE Dynamics

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Abstract. This project benchmarks data-driven and physics-structured models for learning nonlinear ODE dynamics from noisy trajectories. I evaluate baseline MLPs, Neural ODE rollouts, and a structured Lagrangian neural network (LNN) on two course systems: Van der Pol and the simple pendulum. Key findings: (1) the LNN reduces trajectory RMSE by two orders of magnitude over black-box baselines while nearly conserving energy; (2) on single-trajectory Van der Pol fitting, a compact MLP outperforms long-horizon rollout training; (3) deeper networks are not uniformly better due to spectral bias and optimization difficulty.

1. Introduction

Learning dynamics from noisy observations is a central problem in scientific machine learning. Black-box regressors fit data efficiently but may violate dynamical consistency; rollout-based models are physically motivated but accumulate integration error; physics-structured models exploit known invariances at the cost of specialization. This project extends Chapters 5 and 7 homework into a unified benchmark with reproducible code, common metrics, and direct comparison across model families.

Objective. Compare baseline MLPs, Neural ODEs, and a structured LNN in terms of trajectory RMSE, energy conservation, and training cost.

2. Method

Van der Pol ($\dot{z}{=}v,\ \dot{v}{=}\mu(1{-}z^2)v{-}z,\ \mu{=}1$): RK4 on $t\in[0,20]$, $\Delta t{=}0.1$, Gaussian noise $\sigma{=}0.1$. MLP capacity study ($2{\times}32$, $6{\times}32$, $2{\times}128$) and Neural ODE rollout.

Pendulum ($\ddot{\theta}{+}\frac{g}{\ell}\sin\theta{=}0,\ \frac{g}{\ell}{=}9.81$): 80 RK4 trajectories on $t\in[0,10]$ (400 steps), noise $\sigma{=}0.02$, split 56/12/12. One-step baseline MLP with autoregressive rollout, Neural ODE, and structured LNN trained on finite-difference acceleration targets via $\ddot{\theta}_{\text{pred}}{=}-\frac{dV_W}{d\theta}$.

3. Results

Table 1: Trajectory RMSE, parameter count, and training time.

System	Model	St. 1 RMSE	St. 2 RMSE	Params	Time (s)
Van der Pol	Baseline MLP	0.053 (z)	0.239 (v)	1 153	0.4
Van der Pol	Neural ODE	0.188 (z)	0.383 (v)	4 482	23.7
Pendulum	Baseline MLP	0.496 (θ)	1.511 (ω)	4 482	2.7
Pendulum	Neural ODE	0.188 (θ)	0.499 (ω)	4 482	52.8
Pendulum	Structured LNN	**0.006** (θ)	**0.017** (ω)	8 513	12.1

Capacity and spectral bias. On Van der Pol, the compact $2{\times}32$ MLP (1 153 params) outperformed both the deep $6{\times}32$ (RMSE 0.895) and wide $2{\times}128$ (RMSE 0.104) variants. The deep model failed to fit oscillatory structure, consistent with spectral bias and vanishing-gradient effects in deeper tanh networks.

Energy conservation. For pendulum, the LNN rollout shows energy drift std of 0.002 versus 0.553 for Neural ODE, demonstrating that the physics-structured architecture implicitly preserves the system's Hamiltonian structure.

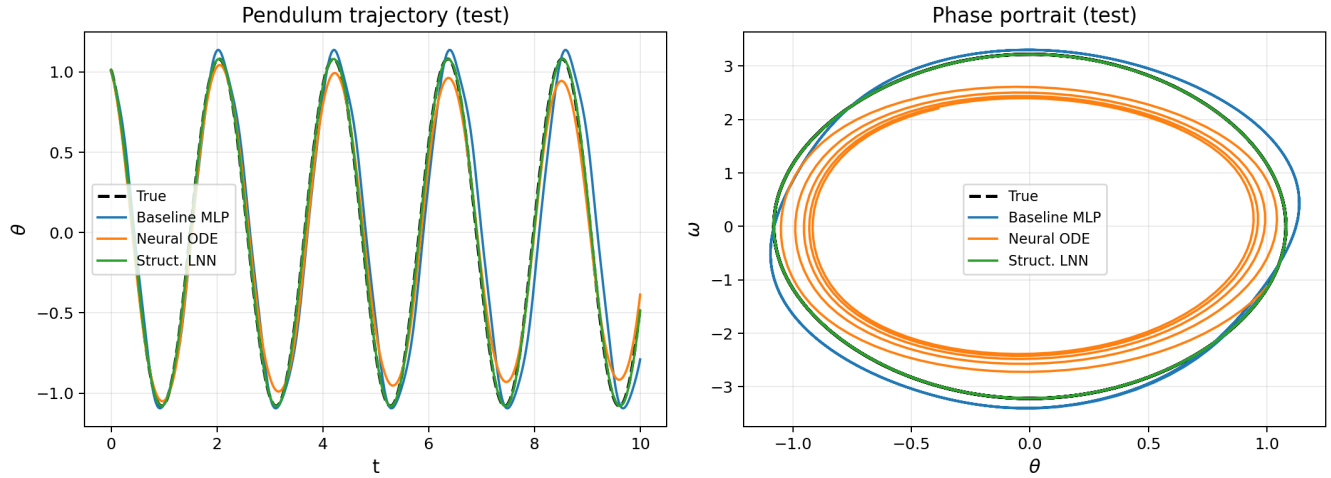

Figure 1: Representative pendulum test trajectory (left) and phase portrait (right). The structured LNN closely tracks the true dynamics while baseline and Neural ODE drift.

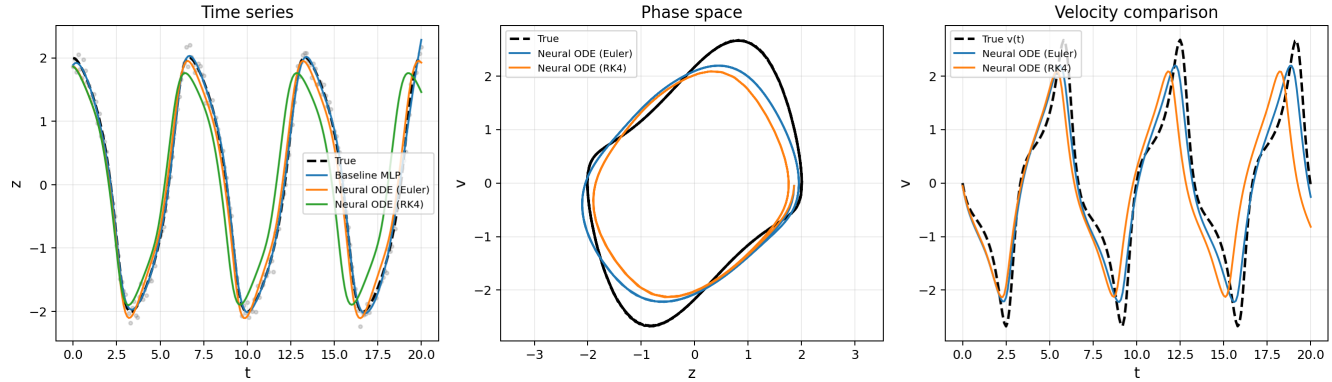

Figure 2: Van der Pol: time series, phase space, and velocity comparisons between baseline MLP, Neural ODE (Euler), and Neural ODE (RK4 evaluation).

4. Discussion

Three findings stand out. First, model capacity does not automatically improve fitting: the compact MLP outperformed larger variants on Van der Pol, highlighting that architecture search matters. Second, integration error compounds: the Neural ODE trained with Euler rollout underperformed direct regression on a single trajectory, though it better captured phase-space structure. Third, when correct physics structure is available, the inductive bias is decisive: the pendulum LNN reduced RMSE by nearly 100× and preserved energy conservation.

Limitations. Van der Pol uses one initial condition and first-order Euler for Neural ODE training; both choices disadvantage rollout models. The LNN assumes known kinetic energy form, which may not generalize to systems without such decomposition.

Future work. Higher-order differentiable integrators for Neural ODE training, PINN-style residual constraints, and extension to chaotic systems (Lorenz, double pendulum).

Reproducibility. All code: https://github.com/Xuanyou-Liu/ELE_ENG-395-Final-Project. Run `bash scripts/reproduce`